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Proof-of-Work versus Proof-of-Stake: A Comparative Economic Analysis

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1 Big picture

- **Question.** Which blockchain protocol is more secure in equilibrium: Proof-of-Work (PoW) or Proof-of-Stake (PoS)? How does the answer depend on adoption, scale, DeFi opportunities, validation costs, and attacker incentives?
- **Main contribution.** The paper builds a unified economic model in which users choose whether to adopt a blockchain, how to allocate crypto wealth between validation and DeFi, and whether to pay fees to avoid transaction delay. An attacker chooses whether to disrupt the chain. The model solves equilibrium adoption, validation investment, fees, asset value, and security under both PoW and PoS.
- **Key comparative insight.** PoW and PoS differ in the cost of attack and the opportunity cost of honest validation. PoW validation requires mining hardware and energy, which depreciate; PoS validation requires holding the native cryptoasset dormant as stake.
- **Main theoretical result.** Under a general condition, PoS generates higher equilibrium security than an otherwise equivalent PoW blockchain. The condition compares the cost advantage of PoS validation with the possibility that a PoS attacker can attack with less than a majority of total stake and perhaps borrow native cryptoasset.
- **Main calibration result.** Using realistic parameter estimates, mining hardware depreciates much faster than the relevant DeFi opportunity return. Therefore, PoS is more secure than PoW unless a PoS attacker can borrow several hundred percent more native cryptoasset capital than a PoW attacker can borrow.
- **Scale result.** At high scale, the PoS security advantage becomes more salient. Scaling can make PoW become fully insecure at lower scale than PoS. Conversely, scaling can make PoS fully secure at lower scale than PoW when attacker benefits are sufficiently low.
- **Mechanism for scaling risk.** Higher scale reduces congestion and user fees. Lower fees benefit users but reduce validator fee revenue. If attacker benefits are large, lower validation revenue can undermine security.
- **Mechanism for scaling benefits.** Higher scale also increases adoption by lowering user fees. More adoption raises demand for the native cryptoasset and increases the value of block rewards. When attacker benefits are small, this adoption effect can strengthen security.
- **Economic takeaway.** Blockchain security is not purely technical. It is an equilibrium outcome of validator incentives, user adoption, transaction congestion, DeFi opportunity costs, asset values, and attacker incentives. Under current real-world conditions, PoS has an economically important security advantage over PoW.

2 Model primitives and measurement

2.1 Environment and timing

- Time is discrete and infinite, $t \in \mathbb{N}$.
- The economy has overlapping generations. Each generation has a unit mass of agents indexed by $i \in [0, 1]$.
- Agent (i, t) lives for three periods: t , $t + 1$, and $t + 2$.
- The agent receives one unit of endowment only at t and consumes only at $t + 2$.
- The outside option gives gross payoff $1 + \sigma$ with $\sigma \geq 0$.
- Alternatively, the agent can adopt the blockchain by buying the cryptoasset at t , investing the cryptoasset during $t + 1$, and liquidating cryptoasset holdings for consumption at $t + 2$.

Interpretation: This overlapping-generations structure lets the model track three distinct decisions: entry/adoption, investment between validation and DeFi, and final liquidation through a congested blockchain.

2.2 User utility and blockchain adoption

The agent chooses whether to use the blockchain or the outside option:

$$U_{i,t}^p = \max\{\bar{U}_{i,t}^p, 1 + \sigma\}, \quad p \in \{\text{PoW}, \text{PoS}\}, \quad (1)$$

where $\bar{U}_{i,t}^p$ is expected utility from adopting a blockchain with protocol p .

If the agent adopts, she allocates wealth between validation and DeFi:

$$\omega_{i,t}^p = \arg \max_{\omega \in [0,1]} \omega E_t[R_t^p] + (1 - \omega) E_t[R_t^{\text{DeFi}}]. \quad (2)$$

If the chain survives, the gross blockchain return is

$$\Phi_{i,t}^p = \omega_{i,t}^p R_t^p + (1 - \omega_{i,t}^p) R_t^{\text{DeFi}}. \quad (3)$$

Interpretation: Adoption is endogenous. Higher expected returns, lower waiting costs, lower fees, and higher survival probability all make adoption more attractive.

2.3 DeFi application

A DeFi investment yields a per-period gross cryptoasset return $1 + \psi$. Because the agent buys the cryptoasset at t and sells at $t + 2$, the lifetime DeFi return is

$$R_t^{\text{DeFi}} = \frac{P_{t+2}}{P_t} (1 + \psi). \quad (4)$$

The paper assumes $\psi > \sigma$ so that blockchain investment can be attractive relative to the outside option.

Interpretation: The DeFi return is the opportunity cost of validation. In PoS, staking means holding the native cryptoasset dormant; therefore, DeFi yields are a natural benchmark for the opportunity cost of staking.

2.4 Congestion, waiting costs, and fees

The blockchain has finite transaction capacity. Users who sell cryptoassets at $t + 2$ face waiting costs. Each user has wait disutility $c_{i,t}$ drawn from a distribution G on $[0, \infty)$.

A user can pay a fee f to receive higher transaction priority. Let $W^p(f, f_{-(i,t)})$ be expected wait time. The fee problem is

$$f_{i,t}^p = \arg \max_{f \geq 0} \{ \Phi_{i,t}^p - f - c_{i,t} W^p(f, f_{-(i,t)}) \}. \quad (5)$$

Interpretation: Fees are not exogenous. They are equilibrium payments for priority processing. Users with high waiting disutility pay more, so fees are endogenous validator revenue.

2.5 PoW validation

In PoW, validation is mining. Miners buy mining equipment and energy to generate hash power. Let H_{t+1} be aggregate user hash power, κ_h be energy cost per hash-power unit, and δ be depreciation of mining equipment.

The lifetime return to mining has the form

$$R_t^{\text{PoW}} = \frac{H_{t+1}(1 - \delta) + \left(B_{t+1} + \frac{\int f_{i,t-1}^{\text{PoW}} dG(c_{i,t-1})}{P_{t+1}^{\text{PoW}}} \right) P_{t+2}^{\text{PoW}}}{H_{t+1}(1 + \kappa_h)}. \quad (6)$$

Interpretation: PoW validator revenue comes from block rewards and user fees, but mining equipment depreciates and energy must be paid. These costs are important for comparing honest validation to attack costs.

2.6 PoS validation

In PoS, validation is staking. A user stakes cryptoasset and receives a pro rata share of block rewards and fees. Let S_{t+1} be total staked cryptoasset. The return to staking is

$$R_t^{\text{PoS}} = \frac{S_{t+1} P_{t+2}^{\text{PoS}} + \left(B_{t+1} + \frac{\int f_{i,t-1}^{\text{PoS}} dG(c_{i,t-1})}{P_{t+1}^{\text{PoS}}} \right) P_{t+2}^{\text{PoS}}}{S_{t+1} P_t^{\text{PoS}}}. \quad (7)$$

Interpretation: Unlike PoW mining equipment, staked cryptoasset is returned to the validator if the blockchain survives. This is why the opportunity cost of staking is central to PoS security.

2.7 Attacker benefit and disruption

Each period, an attacker draws a private benefit $\Theta_t \sim U[0, \bar{\Theta}]$ from successfully disrupting the blockchain. The attacker chooses resources $A_t \geq 0$ to maximize

$$\max_{A_t \geq 0} \Theta_t \nu_{t+1}^p(A_t) - A_t, \quad (8)$$

where $\nu_{t+1}^p(A_t)$ is the probability of successful attack under protocol p .

Interpretation: Security is the probability that the chain survives. Higher validator investment raises the cost of disruption and therefore increases security.

2.8 PoW attacks

A PoW attack succeeds if the attacker obtains at least as much computational power as honest miners. Since hash power costs $1 + \kappa_h$ per unit, the success condition is

$$A_t \geq H_{t+1}(1 + \kappa_h). \quad (9)$$

Thus

$$\nu_{t+1}^{\text{PoW}}(A_t) = \begin{cases} 1, & A_t \geq H_{t+1}(1 + \kappa_h), \\ 0, & A_t < H_{t+1}(1 + \kappa_h). \end{cases} \quad (10)$$

Interpretation: In PoW, the attack cost is tied to honest hash-power investment. If mining investment falls, the cost of attack falls.

2.9 PoS attacks

A PoS attack succeeds if the attacker stakes enough cryptoasset to block supermajority approval. Let $\eta \geq 1/2$ be the required supermajority. Then the attacker needs at least a fraction $1 - \eta$ of overall stake.

The paper allows the PoS attacker to borrow native cryptoasset. Let $\lambda \geq 1$ be relative native-cryptoasset leverage. If the attacker spends A_t , she can acquire $\lambda A_t / P_t^{\text{PoS}}$ coins. The attack condition is

$$\frac{\lambda A_t}{P_t^{\text{PoS}}} \geq (1 - \eta) \left(S_{t+1} + \frac{\lambda A_t}{P_t^{\text{PoS}}} \right). \quad (11)$$

Interpretation: This formulation deliberately allows a potential disadvantage for PoS: a PoS attacker may need less than majority stake and may be able to borrow the native cryptoasset. Even with this conservative assumption, calibrated results favor PoS.

3 Equilibrium and theoretical results

3.1 Lemma 1: optimal fees

In equilibrium, fees are increasing in users' waiting costs:

$$f^p(c) = \begin{cases} \frac{1}{\Gamma} \int_0^c x dG(x), & c < c^p, \\ 0, & c \geq c^p, \end{cases} \quad (12)$$

where Γ denotes blockchain scale and c^p is the adoption cutoff under protocol p .

Interpretation: Users with higher waiting disutility pay higher fees. Users with $c \geq c^p$ do not adopt the blockchain and therefore pay zero fees. A larger scale Γ reduces equilibrium fees.

3.2 Adoption cutoff

Each protocol has an adoption cutoff c^p . Users adopt if and only if

$$c_{i,t} < c^p. \quad (13)$$

Equilibrium adoption is therefore $G(c^p)$.

Interpretation: The cutoff summarizes user-side blockchain demand. Higher security, lower fees, and higher returns all raise c^p and therefore increase adoption.

3.3 Proposition 1: PoW equilibrium

A PoW equilibrium is characterized by:

- adoption cutoff c^{PoW} ;
- market value $M^{\text{PoW}}(c^{\text{PoW}})$;
- mining return $R^{\text{PoW}}(c^{\text{PoW}})$;
- fraction of wealth allocated to mining $\omega^{\text{PoW}}(c^{\text{PoW}})$;
- equilibrium hash power $H^*(c^{\text{PoW}})$;
- survival probability $\pi^{\text{PoW}}(c^{\text{PoW}})$.

The key validation-investment relation is

$$H^*(c^{\text{PoW}})(1 + \kappa_h) = \omega^{\text{PoW}}(c^{\text{PoW}})G(c^{\text{PoW}}). \quad (14)$$

The survival probability is

$$\pi^{\text{PoW}}(c^{\text{PoW}}) = \min\{\hat{\pi}^{\text{PoW}}(c^{\text{PoW}}), 1\}. \quad (15)$$

Interpretation: In PoW, security depends on user investment into hash power. Mining hardware depreciation and energy costs lower the private return to validation and can weaken security.

3.4 Proposition 2: PoS equilibrium

A PoS equilibrium is characterized by:

- adoption cutoff c^{PoS} ;
- market value $M^{\text{PoS}}(c^{\text{PoS}})$;

- staking return $R^{\text{PoS}}(c^{\text{PoS}})$;
- fraction of wealth allocated to staking $\omega^{\text{PoS}}(c^{\text{PoS}})$;
- total staked cryptoasset value $S^*(c^{\text{PoS}})$;
- survival probability $\pi^{\text{PoS}}(c^{\text{PoS}})$.

The key validation-investment relation is

$$S^*(c^{\text{PoS}}) = \omega^{\text{PoS}}(c^{\text{PoS}})G(c^{\text{PoS}}). \quad (16)$$

The survival probability is

$$\pi^{\text{PoS}}(c^{\text{PoS}}) = \min\{\hat{\pi}^{\text{PoS}}(c^{\text{PoS}}), 1\}. \quad (17)$$

Interpretation: In PoS, validation investment is native cryptoasset staked by users. Because the principal is returned if the chain survives, the main cost of staking is the opportunity cost relative to DeFi.

3.5 Comparing PoW and PoS

The general comparison can be summarized as:

$$\pi^{\text{PoS}} \geq \pi^{\text{PoW}} \iff \text{PoS attack-cost disadvantage is outweighed by PoS validation-cost advantage.} \quad (18)$$

- PoS can have an attack-cost disadvantage because a successful attack may require less than a majority stake and because the attacker may borrow native cryptoasset.
- PoS has a validation-cost advantage because staking does not require depreciating mining hardware and energy expenditure.
- PoW validation can be expensive and depreciating, which lowers honest validation investment relative to an otherwise comparable PoS chain.

Interpretation: The paper’s main comparison is an equilibrium comparison, not a purely mechanical comparison of 51% attacks versus stake attacks.

3.6 Proposition 3: general comparison

The paper derives a general condition that determines when PoS security exceeds PoW security. The condition can be written conceptually as

$$\alpha \geq \zeta \implies \pi^{\text{PoS}} \geq \pi^{\text{PoW}}, \quad (19)$$

where α summarizes the effective attacker-cost parameter for PoS and ζ summarizes the validation-cost advantage that PoS must overcome.

Interpretation: Proposition 3 provides the theoretical comparison. The rest of the paper calibrates the relevant parameters and studies how the comparison behaves as scale changes.

3.7 Proposition 4: lower-dimensional sufficient condition

The paper reduces the comparison to a condition involving three key real-world parameters:

- η : the supermajority required by the PoS protocol;
- λ : the relative ability of a PoS attacker to borrow native cryptoasset;
- δ_{1Y} and ψ_{1Y} : annual mining-hardware depreciation and annual DeFi return.

If $\eta \leq 2/3$, $\delta_{1Y} > \psi_{1Y}$, and

$$\lambda \leq \frac{1}{2} \left(1 + \frac{\log(1 + \delta_T)}{\log(1 + \psi_T)} \right), \quad (20)$$

then PoS generates higher equilibrium security than PoW. Because

$$\frac{\log(1 + \delta_T)}{\log(1 + \psi_T)} = \frac{\log(1 + \delta_{1Y})}{\log(1 + \psi_{1Y})}, \quad (21)$$

the condition is invariant to the horizon T .

Interpretation: This is the paper’s calibration-friendly result. It says the authors only need estimates of mining-hardware depreciation, DeFi returns, and relative PoS-native borrowing ability.

3.8 Corollary 1: real-world calibration

Using conservative estimates:

$$\delta_{1Y} \geq 46.5\%, \quad \psi_{1Y} \leq 4.85\%, \quad (22)$$

PoS is more secure than PoW whenever

$$\lambda \leq 453\%. \quad (23)$$

Using less conservative estimates:

$$\delta_{1Y} \geq 55\%, \quad \psi_{1Y} \leq 4.32\%, \quad (24)$$

PoS is more secure whenever

$$\lambda \leq 568\%. \quad (25)$$

Interpretation: A PoS attacker would need to borrow several hundred percent more native cryptoasset capital than a PoW attacker to overturn the PoS security advantage. The authors argue that current crypto lending markets make such leverage unrealistic.

4 Scale, security, and adoption

4.1 Why scale matters

Let Γ denote blockchain scale. Scale affects the equilibrium through congestion and fees:

$$f^p(c) = \frac{1}{\Gamma} \int_0^c x dG(x). \quad (26)$$

Thus, as Γ increases, fees fall.

- Lower fees increase user adoption because users lose less surplus to congestion costs.
- Lower fees reduce validator fee revenue, which can reduce validation investment.
- Higher adoption raises cryptoasset demand and increases the value of block rewards.

Interpretation: Scaling has two opposite effects. It can increase adoption and block-reward value, but it can also reduce fee revenue for validators.

4.2 Full insecurity at high scale

Define the set of maximal attacker benefits that generate full insecurity for high-scale blockchains:

$$\mathcal{S}_0^p = \{\bar{\Theta} > 0 : \exists \Gamma^p > 0 \text{ such that } \pi^p = 0 \text{ for all } \Gamma > \Gamma^p\}. \quad (27)$$

Interpretation: This set describes when scaling eventually destroys all security.

4.3 Proposition 5: high-scale PoW is insecure for a larger attacker-benefit set

Under realistic parameters $\alpha \geq \zeta$,

$$\mathcal{S}_0^{\text{PoS}} \subseteq \mathcal{S}_0^{\text{PoW}}. \quad (28)$$

Interpretation: Scaling can render PoW fully insecure in cases where PoS remains secure. But if scaling renders PoS fully insecure, it also renders PoW fully insecure.

4.4 Proposition 6: PoW becomes fully insecure at lower scale

For a fixed maximal attacker benefit $\bar{\Theta}$, define

$$\Gamma_0^p = \sup\{\Gamma > 0 : \pi^p > 0\}. \quad (29)$$

Then

$$\Gamma_0^{\text{PoS}} \geq \Gamma_0^{\text{PoW}}. \quad (30)$$

Interpretation: If scaling eventually causes full insecurity, PoW reaches that failure point at a lower scale than PoS.

4.5 Full security at high scale

Define the set of maximal attacker benefits that generate full security for high-scale blockchains:

$$\mathcal{S}_1^p = \{\bar{\Theta} > 0 : \exists \Gamma^p > 0 \text{ such that } \pi^p = 1 \text{ for all } \Gamma > \Gamma^p\}. \quad (31)$$

Interpretation: This set describes when scaling eventually makes a blockchain fully secure.

4.6 Proposition 7: high-scale PoS is fully secure for a larger attacker-benefit set

Under realistic parameters $\alpha \geq \zeta$,

$$\mathcal{S}_1^{\text{PoW}} \subseteq \mathcal{S}_1^{\text{PoS}}. \quad (32)$$

Interpretation: Scaling generates full security for PoS in cases where it does not generate full security for PoW.

4.7 Proposition 8: PoS becomes fully secure at lower scale

For a fixed maximal attacker benefit $\bar{\Theta}$, define

$$\Gamma_1^p = \sup\{\Gamma > 0 : \pi^p < 1\}. \quad (33)$$

Then

$$\Gamma_1^{\text{PoS}} \leq \Gamma_1^{\text{PoW}}. \quad (34)$$

Interpretation: If scaling eventually produces full security, PoS reaches full security at a lower scale than PoW.

4.8 Overall scale lesson

- When attacker benefits are large, scaling can weaken security by reducing fee revenue and validation incentives.
- This negative effect is more urgent for PoW.
- When attacker benefits are small, scaling can strengthen security by increasing adoption and cryptoasset value.
- This positive effect is more immediate for PoS.

Interpretation: Scale does not have one universal effect. Its security effect depends on attacker incentives, but the comparison favors PoS in both directions under realistic parameters.

5 Empirical parameter evidence and interpretation

5.1 Mining hardware depreciation

The authors estimate annual depreciation rates for ASIC mining equipment using public data and prior academic studies.

- Theblock.co ASIC machine price indexes imply average annual depreciation between 60.4% and 66.38%.
- Prat and Walter (2021) imply 47.7%–61.3%.
- De Vries and Stoll (2021) imply 46.5%–89%.
- Stinner (2021) implies 61.45%.

Interpretation: PoW mining equipment loses value quickly. This high depreciation raises the opportunity cost of honest mining and weakens PoW relative to PoS.

5.2 DeFi and staking returns

The relevant DeFi return is the return on a risk-equivalent alternative to staking. The paper focuses on ETH staking and ETH lending.

- Rolling annual staked ETH returns from Compass FT average 4.32%.
- The minimum is 3.73% and the maximum is 4.85%.
- AAVE ETH lending averages 1.66% from 2023–2024.
- Academic DeFi lending estimates are even lower in ETH lending data.

Interpretation: The relevant PoS opportunity cost is much lower than PoW mining hardware depreciation. This is why PoS tends to have a large validation-cost advantage.

5.3 Relative PoS attacker leverage

The model allows a PoS attacker to borrow native cryptoasset through $\lambda \geq 1$. Current lending markets are largely overcollateralized, so the authors argue that $\lambda \approx 1$ in practice.

Interpretation: This is important because the calibration allows PoS to be disadvantaged by leverage. Even then, PoS remains more secure unless attacker leverage is implausibly large.

6 Tables and figures

The paper does not rely on plotted figures. The central visual material is the parameter tables, which are directly cropped from the paper and included below.

6.1 Tables 1 and 2: depreciation and staking returns

Read:

- Table 1 reports annual depreciation estimates for ASIC mining equipment.
- Across sources, depreciation is always high, ranging at least from 46.5% to 89%.
- Table 2 reports rolling annual staked ETH returns from Compass FT data.
- The average annual staking return is 4.32%, with range 3.73%–4.85%.

Economic message: Mining equipment depreciates far faster than ETH staking/DeFi returns. This is the numerical foundation for the PoS security advantage.

Table 1 Estimates for annual depreciation rates of ASIC mining equipment (see Appendix A)	
Data source	Estimate for average annual depreciation rate δ_{1Y}
theblock.co	60.4% - 66.38%
Academic source	Estimate for average annual depreciation rate δ_{1Y}
Prat and Walter 2021	47.7% - 61.3%
De Vries and Stoll 2021	46.5% - 89%
Sinner 2021	61.45%

Table 1: ASIC mining equipment depreciation

Table 2 Summary statistics of rolling annual staked ETH returns using Compass FT data					
Variable	Average	Median	Min	Max	SD
Rolling staked ETH return (Ann)	4.32%	4.33%	3.73%	4.85%	.0031

Table 2: Rolling annual staked ETH returns

6.2 Table 3: DeFi lending returns

Read:

- Table 3 reports additional DeFi ETH lending estimates.
- AAVE ETH lending averages 1.66% over 2023–2024.
- Chaudhary, Kozhan, and Vishwanath-Natraj (2023) report ETH DeFi lending average 0.13% and maximum 2.28%.

Economic message: Robust DeFi return estimates remain well below mining hardware depreciation, so the conclusion that PoS is more secure than PoW is robust to alternative DeFi measures.

Table 3 Annualized DeFi ETH lending return, ψ (see Appendix A)	
Source	Annualized DeFi ETH lending return ψ
AAVE	1.66% average (2023-2024)
Chaudhary, Kozhan, and Vishwanath-Natraj 2023	0.13% average (max 2.28%)

6.3 Appendix Tables A1 and A2: additional parameter evidence

Read:

- Table A1 decomposes ASIC price changes by efficiency tier from 2021 to 2023.
- All efficiency tiers show large annual price declines.
- Table A2 reports additional DeFi lending return estimates across AAVE, Compound, and academic studies.
- The appendix reinforces the same parameter conclusion: mining depreciation is much larger than DeFi returns.

Economic message: The appendix shows that the calibration result is not driven by one data source. Across multiple measures, the sufficient condition for PoS superiority remains satisfied.

Table A1
Average Price Index change YoY from 2021 to 2023 by efficiency tier

Efficiency tier	Price change '21-'22	Price change '22-'23	Price change per year avg. '21-'23
< 25 J/TH	-61.36%	-60.47%	-60.92%
25-38 J/TH	-61.04%	-59.77%	-60.4%
38 J/TH	-67.53%	-65.22%	-66.38%

Table A1: ASIC price index changes

Table A2
Estimates for annual DeFi return, ψ (see Appendix A)

Data source	Estimate for annual DeFi return ψ	Application
AAVE	1.66% average (2023-2024)	ETH DeFi lending (AAVE)
Academic source	Estimate for annual DeFi return ψ	DeFi application
Chaudhary, Kozhan, and Vishwanath-Natraj 2023	0.13% average (max 2.28%)	ETH DeFi lending (AAVE)
Chaudhary, Kozhan, and Vishwanath-Natraj 2023	2.832% average (max 8.957%)	USDC DeFi lending (AAVE)
Park and Stinner 2023	4.8% average	All assets (AAVE)
Park and Stinner 2023	1.2% average	All assets (Compound)

Table A2: DeFi return estimates

7 Short summary

- The paper compares PoW and PoS security in one unified equilibrium model.
- Users choose whether to adopt the blockchain, how to allocate cryptoasset between validation and DeFi, and how much to pay in fees to avoid delay.
- Validators earn block rewards and fees.
- Attackers draw random benefits from disruption and attack only when disruption payoff exceeds attack cost.
- PoW attack cost is tied to the cost of acquiring enough hash power.
- PoS attack cost is tied to acquiring enough stake to block supermajority approval.
- PoS may have an attack-cost disadvantage because attack can require less than majority stake and because the attacker may borrow native cryptoasset.
- PoS has a validation-cost advantage because staking does not require depreciating mining hardware.
- A lower-dimensional condition shows that PoS is more secure when mining depreciation is sufficiently high relative to DeFi returns and PoS attacker leverage is not too large.
- Real-world estimates strongly satisfy this condition.
- PoS remains more secure unless a PoS attacker can borrow several hundred percent more native cryptoasset than a PoW attacker.
- Scaling can either harm or improve security, but high-scale comparisons favor PoS.
- PoW becomes fully insecure at lower scale than PoS when scaling is harmful.
- PoS becomes fully secure at lower scale than PoW when scaling is beneficial.

8 Conclusion

8.1 Core conclusion

John, Rivera, and Saleh show that blockchain security is an equilibrium economic object. It depends on validation incentives, user adoption, congestion fees, DeFi opportunity costs, asset values, and attacker benefits. Under realistic parameter values, PoS blockchains are more secure than otherwise equivalent PoW blockchains.

8.2 Economic intuition

The intuition is that PoW validators must invest in costly depreciating mining hardware and energy, whereas PoS validators stake the cryptoasset and retain the principal if the chain survives. Because mining hardware

depreciation is much larger than the opportunity return to staking, honest validation is economically less costly under PoS. Although PoS attackers may require less than majority stake and may be able to borrow native cryptoasset, realistic leverage is far below the level needed to eliminate the PoS advantage.

8.3 Takeaway

The paper reframes the PoW-versus-PoS debate as a question of equilibrium incentives rather than pure protocol mechanics. Given current economic conditions, PoS provides a stronger security-adoption equilibrium, and its advantage is especially important for high-scale blockchains.